# Math 1610, Honors Calculus, Fall 2026, Worksheet 6

October 20, 2026

Name _____

Please write only your own name in the space above.

1. Suppose $f$ is differentiable on all of $\mathbb{R}$, that $f(0) = 0$, and that $|f'(x)| \leq 1$ for every $x \in \mathbb{R}$. Show that

$$|f(x)| \leq |x|$$

for every $x \in \mathbb{R}$.